School of Technology, Design and Computer Application
Silver Oak College of Compter Application
Department of Computer Application

A PROJECT PRESENTATION ON

Modern to-do

Submitted By

Patel Mitu (2404070100147)
Salunke Ayushi (2404070100190)

Master of Computer Application

Semester III

Group No: 17

Guide

Ms. Jeenal Panchasara

Submitted To

Department of Computer Application
Silver Oak University, Gota, Ahmedabad, 38248
Academic Year 2025-26

1.INTRODUCTION

The "E-commerce Website" project aims to create a dynamic, user-friendly online platform for buying and selling products. This system will serve as a digital storefront, allowing businesses to showcase their products to a global audience and customers to browse, select, and purchase items conveniently. The platform will be scalable, secure, and intuitive, catering to the growing demands of the online retail market.

2.ABSTRACT

This document outlines the development of a comprehensive e-commerce website. The project will address the challenges faced by traditional retail by providing an efficient, accessible, and secure online shopping experience. Key functionalities will include product browsing, a secure shopping cart, multiple payment options, order tracking, and a robust user management system. The system will be built using modern web technologies to ensure a responsive design and high performance.

3.OBJECTIVE

○ To develop a fully functional e-commerce website that supports the entire online shopping process, from product selection to payment and delivery.

○ To provide a secure and reliable platform for online transactions.

○ To create an intuitive and easy-to-navigate user interface for a seamless shopping experience.

○ To implement a robust backend system for inventory management, order processing, and customer data management.

○ To ensure the website is responsive and accessible on various devices, including desktops, tablets, and smartphones.

4.Problem Definition

The primary problem is the inefficiency and limited reach of traditional brick-and-mortar retail. Businesses are often confined to a specific geographical location, limiting their potential customer base. Similarly, consumers are restricted by store operating hours and physical availability of products. The proposed e-commerce website will solve these issues by providing a 24/7 global marketplace, enabling businesses to expand their reach and offering customers unparalleled convenience and access to a wider variety of products.

5. Existing System

Currently, many small and medium-sized businesses either rely on traditional retail methods or use generic social media platforms for sales. These existing systems often lack essential e-commerce features such as integrated payment gateways, automated inventory management, and secure user data handling. This leads to manual and error-prone processes, a poor customer experience, and security vulnerabilities.

Methodology

- The project follows the **Agile development methodology** with iterative development and testing.

Steps:

1.Requirement Gathering – Identify functional and non-functional requirements.

2.System Design – Define database schema, frontend components, and backend APIs.

3.Frontend Development – HTML, CSS and JavaScript are used.

4.Backend Development – PHP (Laravel) for the backend.

5.Database Integration (MySQL or PostgreSQL) – Store user details, groups, and chat history.

6.Testing & Deployment – Perform unit, integration, and user acceptance testing before deployment.

4.1 Requirement Analysis

This phase involves identifying and documenting all the necessary functionalities and constraints for the e-commerce system.

4.1.1 Software Requirements

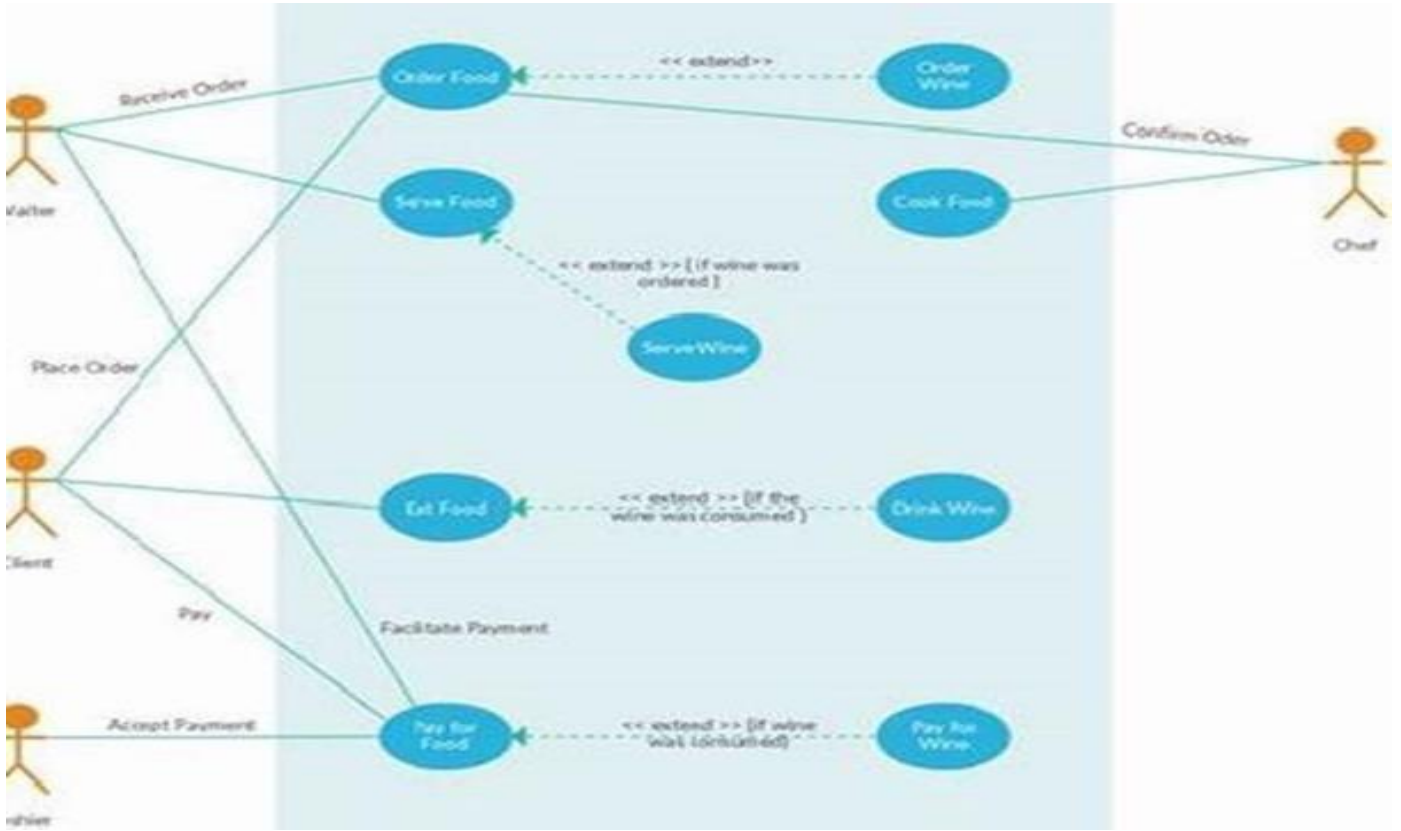
- Operating System: Windows, macOS, or Linux.
- Web Server: Apache or Nginx.
- Database: MySQL or PostgreSQL.
- Programming Languages: HTML, CSS, JavaScript for the frontend; Python (Django/Flask) or PHP (Laravel) for the backend.
- E-commerce Framework: Custom-built or open-source platforms like Magento, WooCommerce, or Shopify.
- Security: SSL/TLS certificate for secure data transmission.
- Payment Gateway: Integration with services like Stripe, PayPal, or a local payment provider.

4.1.2 Hardware Requirements

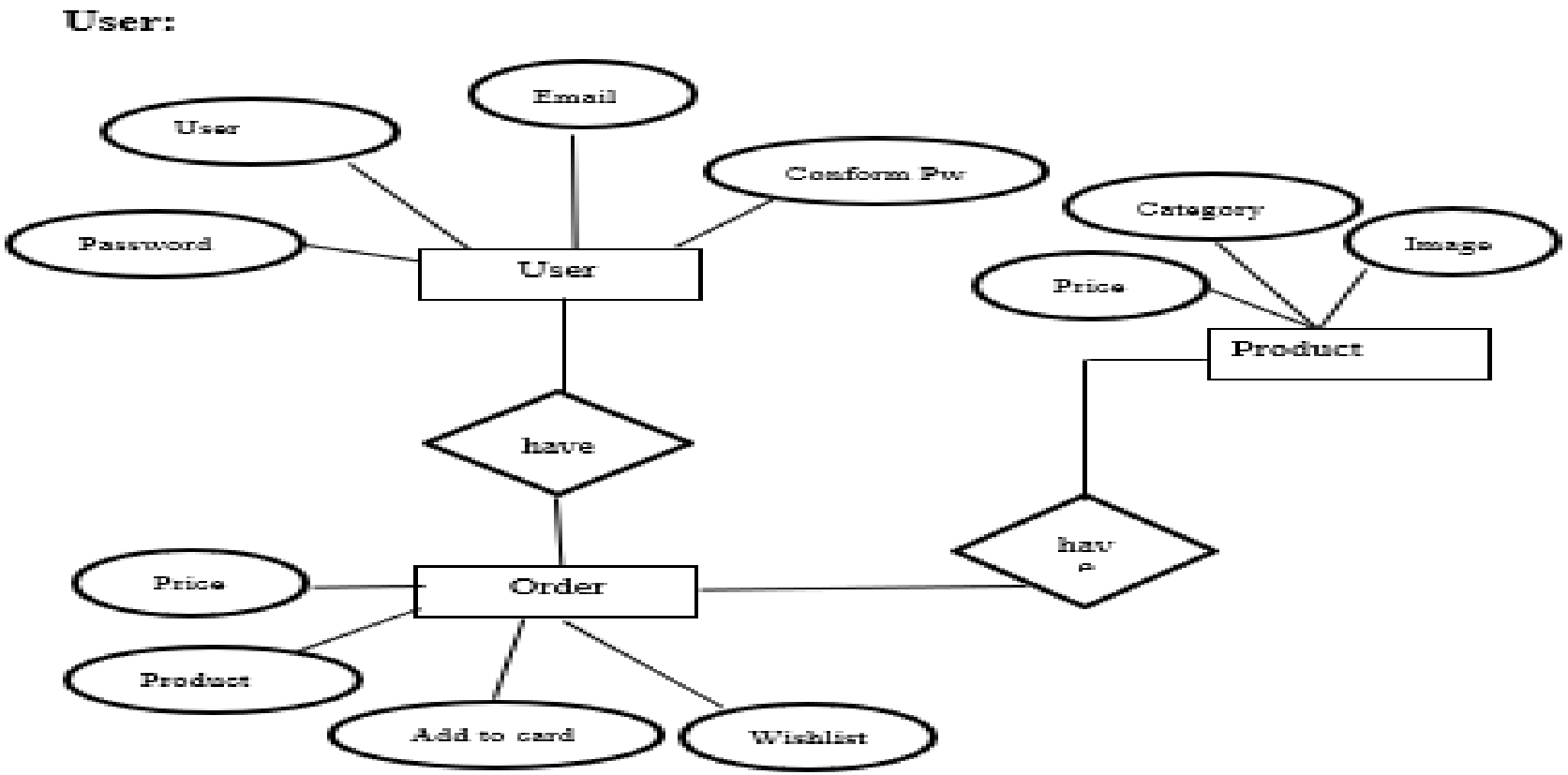
- Server: A dedicated server or cloud hosting service (e.g., AWS, Google Cloud) with a minimum of 8 GB RAM and a multi-core processor to handle user traffic.
- Storage: 500 GB SSD for fast data access and a large volume of product images and data.
- Network: High-speed internet connection for consistent server uptime.

System Design

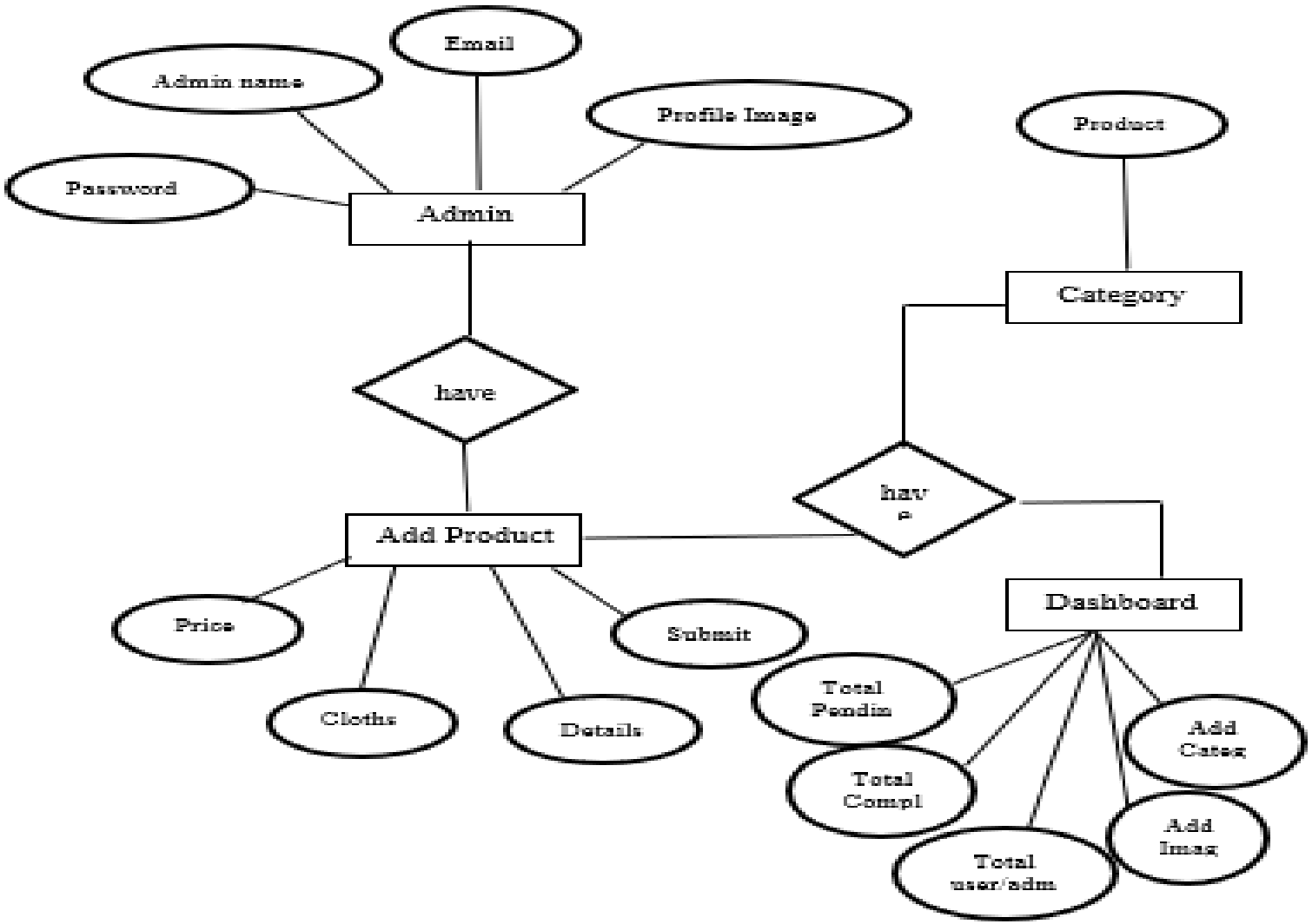
Use Case :



E-R Diagram


User:
Email
User
Conform Pw
Password
User
Category
Image
Price
Product
have
hav
e
Price
Order
Product
Add to card
Wishlist

Admin:

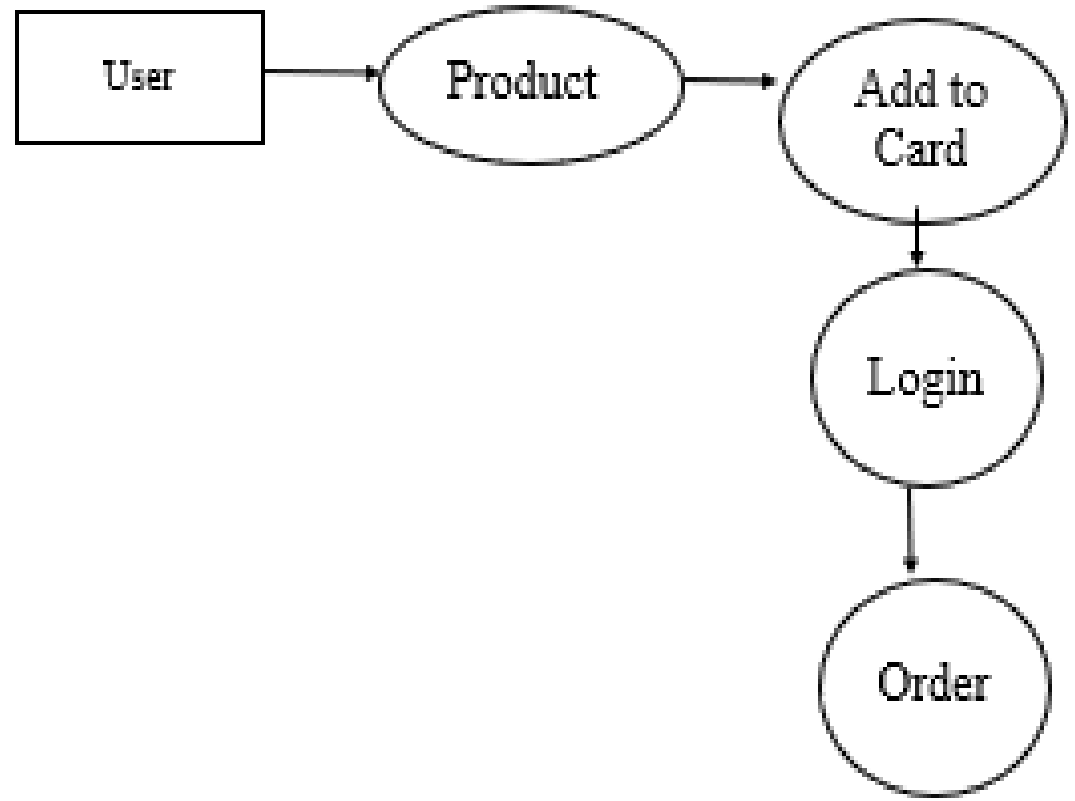

Email
Admin name
Profile Image
Product
Password
Admin
Category
have
hav
e
Add Product
Dashboard
Price
Submit
Total
Pendin
Cloths
Details
Add
Categ
Total
Compl
Add
Imag
Total
user/adm

Data Flow Diagram

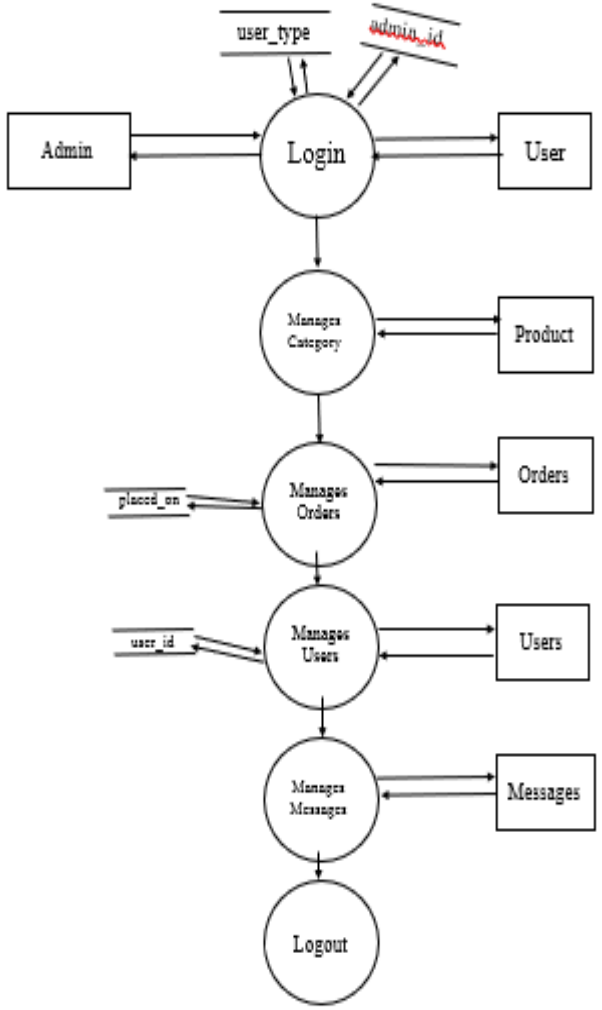
Level 0:

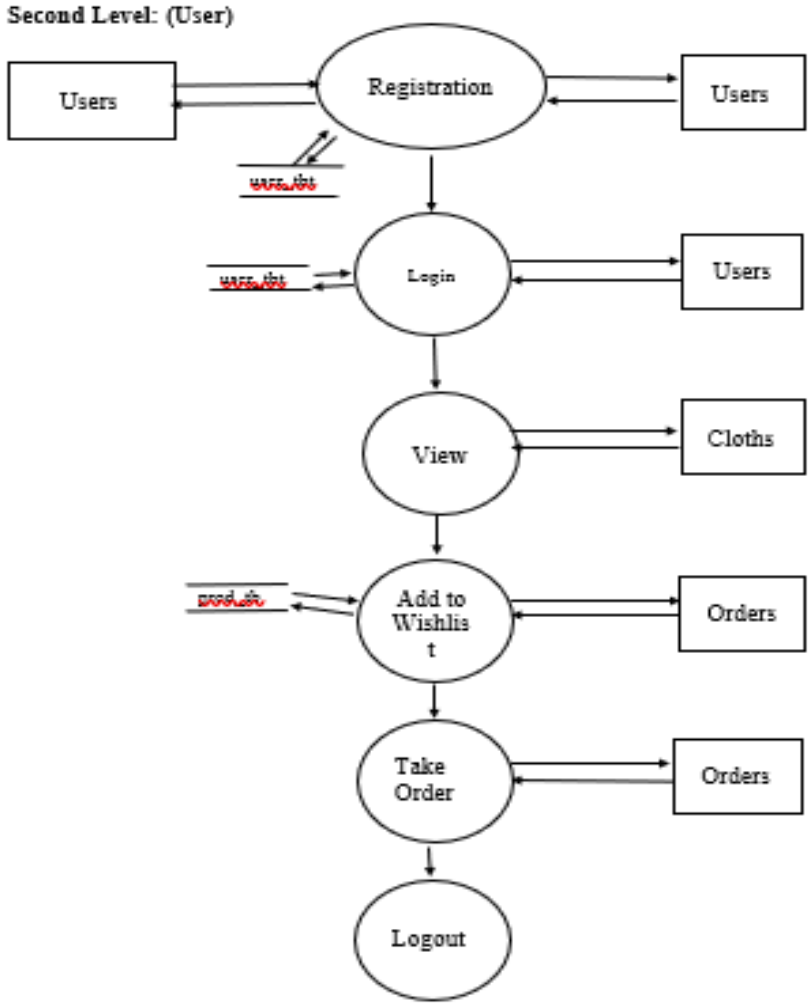


Level 1 DFD for User:

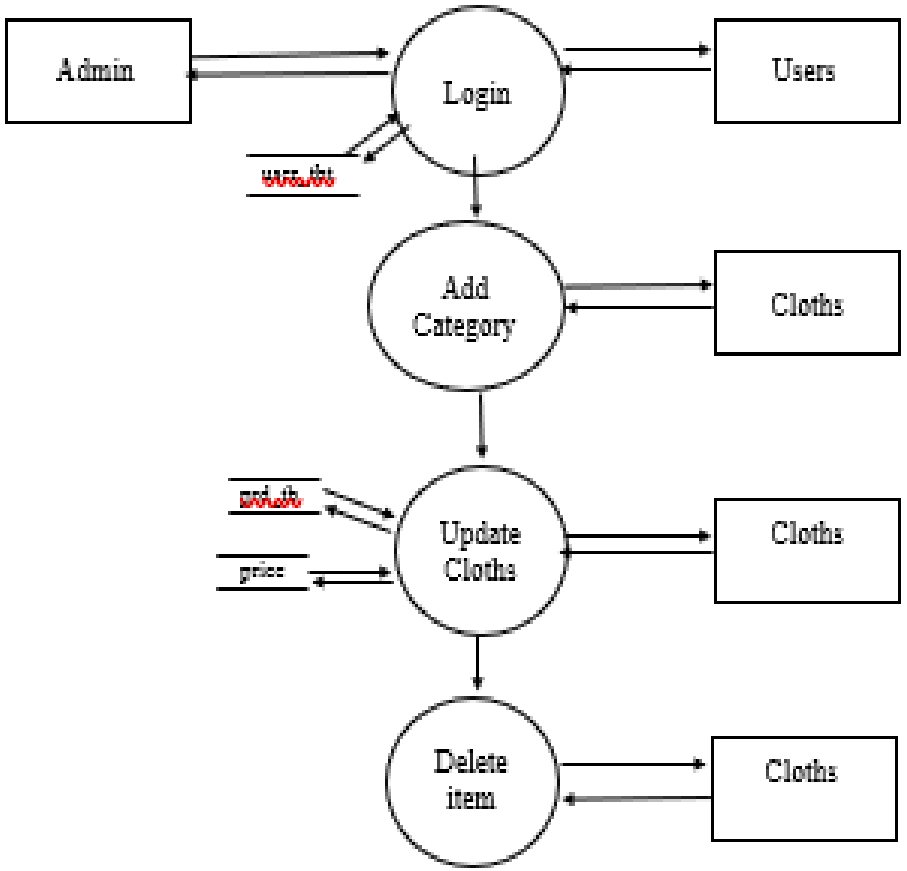


- Level 2 Admin DFD:





Second Level: (Admin)



DATA DICTIONARY

User:

Field	Data Type	Key	Description
user_id	INT AUTO_INCREMENT	PK	Primary key — unique user identifier
Name	VARCHAR(150)		Full name of the user
Email	VARCHAR(255)	UQ	User email (used to login)
password_hash	VARCHAR(255)		Hashed password
Role	ENUM('user','admin')		Role for access control
Phone	VARCHAR(20)		Optional contact number
created_at	DATETIME		Account creation timestamp

Categories:

Field	Data Type	Key	Description
user_id	INT AUTO_INCREMENT	PK	Primary key — unique user identifier
Name	VARCHAR(150)		Full name of the user
Email	VARCHAR(255)	UQ	User email (used to login)
password_hash	VARCHAR(255)		Hashed password
Role	ENUM('user','admin')		Role for access control
Phone	VARCHAR(20)		Optional contact number
created_at	DATETIME		Account creation timestamp

Products:

Field	Data Type	Key	Description
product_id	INT AUTO_INCREMENT	PK	Unique product id
category_id	INT	FK -> categories(category_id)	Product category
Name	VARCHAR(200)		Product title
Slug	VARCHAR(220)	UQ	URL-friendly product name
Description	TEXT		Product detailed description
Price	DECIMAL(10,2)		Unit price
Stock	INT		Number of units available
Image	VARCHAR(255)		Main image path/filename
created_at	DATETIME		Timestamp added

Cart:

Field	Data Type	Key	Description
cart_id	INT AUTO_INCREMENT	PK	Unique cart row id
user_id	INT	FK -> users(user_id)	Owner of the cart row
product_id	INT	FK -> products(product_id)	Product added to cart
product_name	VARCHAR(200)		Snapshot of product name
price	DECIMAL(10,2)		Snapshot of product price
quantity	INT		Quantity selected
added_at	DATETIME		When the item was added

Order :

Field	Data Type	Key	Description
order_id	INT AUTO_INCREMENT	PK	Unique order id
user_id	INT	FK -> users(user_id)	User who placed the order
total_amount	DECIMAL(12,2)		Sum of order items
Status	VARCHAR(50)		e.g. pending, paid, shipped, delivered, cancelled
shipping_address	TEXT		Shipping information snapshot
payment_method	VARCHAR(50)		e.g. card, netbanking, COD
created_at	DATETIME		Order placement time

Messages:

Field	Data Type	Key	Description
message_id	INT AUTO_INCREMENT	PK	Unique message id
user_id	INT NULL	FK -> users(user_id)	If logged in, reference to user
name	VARCHAR(150)		Sender name (for guest messages)
email	VARCHAR(255)		Sender email
subject	VARCHAR(255)		Message subject
message	TEXT		Message body
is_read	TINYINT(1)		0 = unread, 1 = read
created_at	DATETIME		Submission time

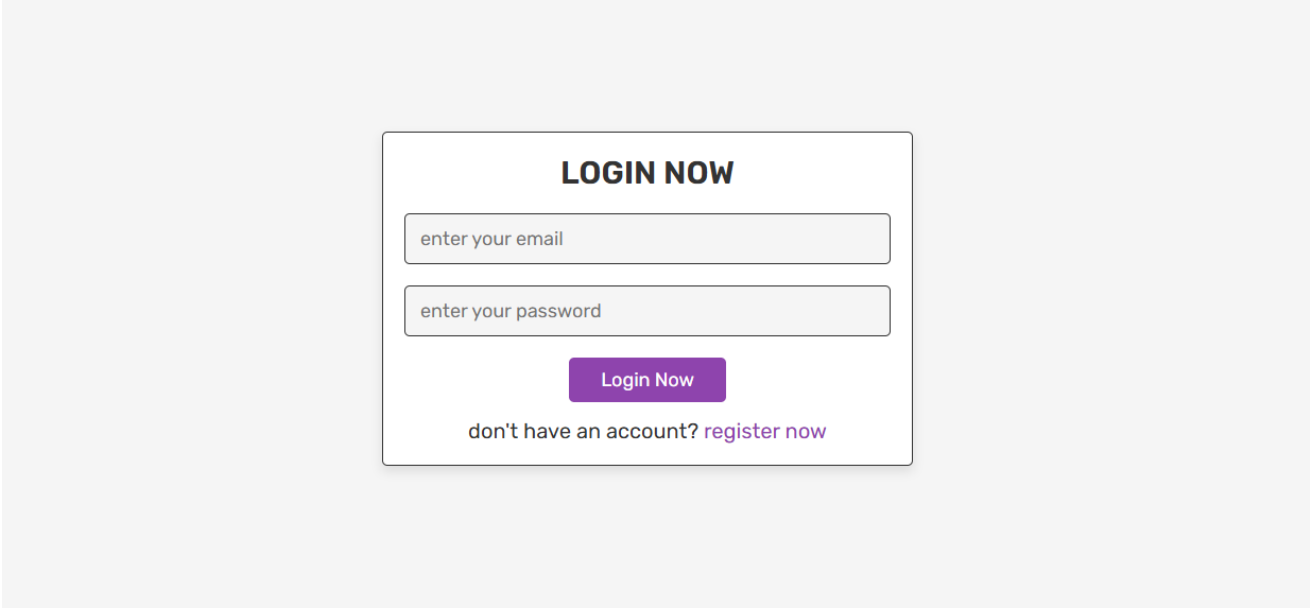
Reviews:

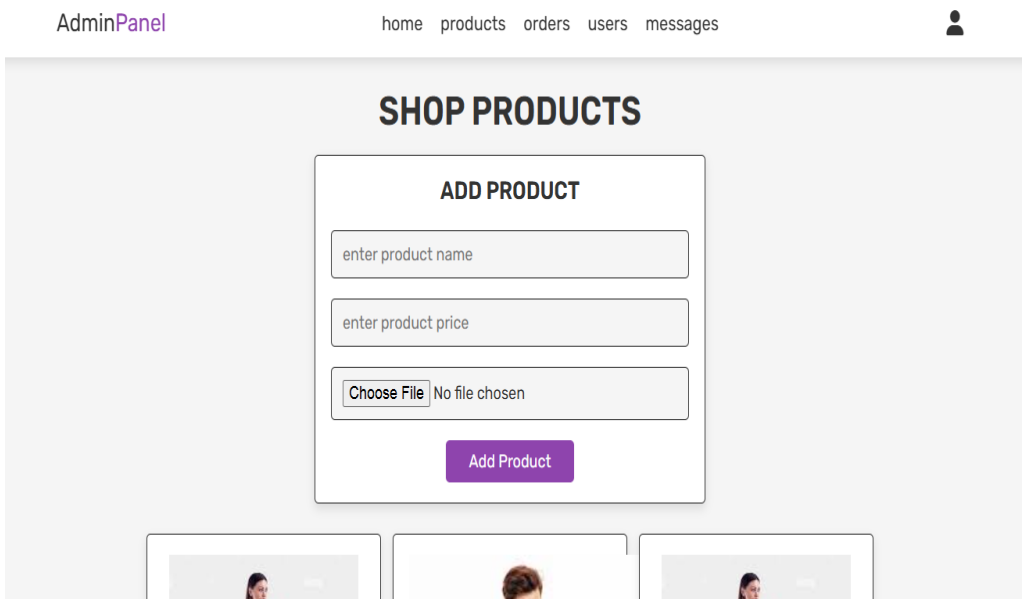
Field	Data Type	Key	Description
review_id	INT AUTO_INCREMENT	PK	Unique review id
product_id	INT	FK -> products(product_id)	Reviewed product
user_id	INT	FK -> users(user_id)	Reviewer
rating	TINYINT		Rating (1-5)
comment	TEXT		Review comment
created_at	DATETIME		When review was posted

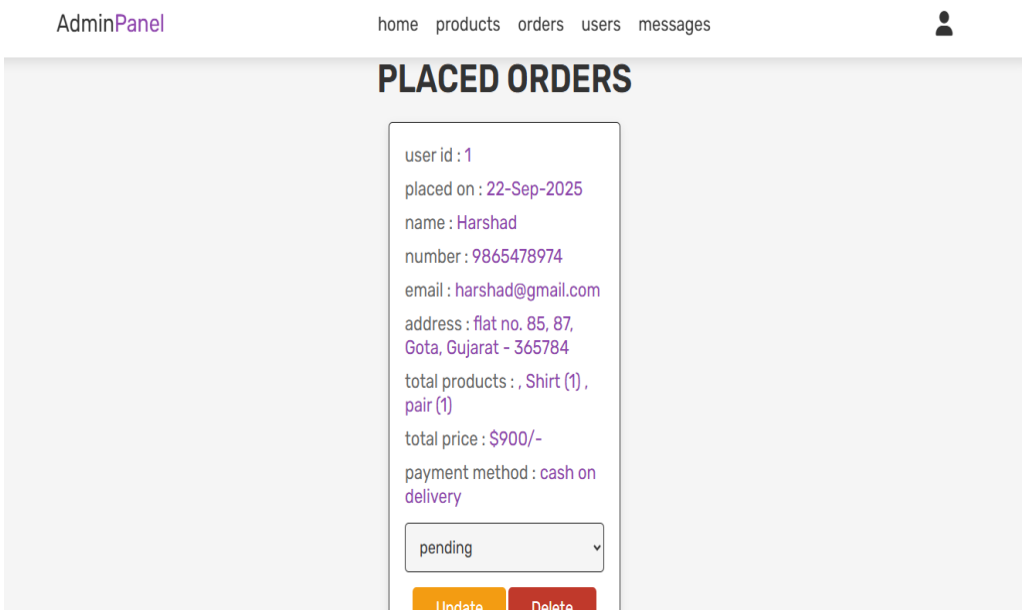
Admin:

Field	Data Type	Key	Description
admin_id	INT AUTO_INCREMENT	PK	Unique admin id
user_id	INT	FK -> users(user_id)	Associated user account
privileges	TEXT		Comma-separated special permissions

Screen Shorts




AdminPanel
home products orders users messages
SHOP PRODUCTS
ADD PRODUCT
enter product name
enter product price
Choose File No file chosen
Add Product


AdminPanel
home products orders users messages
PLACED ORDERS
user id : 1
placed on : 22-Sep-2025
name : Harshad
number : 9865478974
email : harshad@gmail.com
address : flat no. 85, 87, Gota, Gujarat - 365784
total products : , Shirt (1) , pair (1)
total price : $900/-
payment method : cash on delivery
pending
Update
Delete

AdminPanel home products orders users messages

USER ACCOUNTS

user id : 1
username : Harshad
email : harshad@gmail.com
user type : user
Delete User

user id : 2
username : Admin
email : admin@gmail.com
user type : admin
Delete User

AdminPanel home products orders users messages



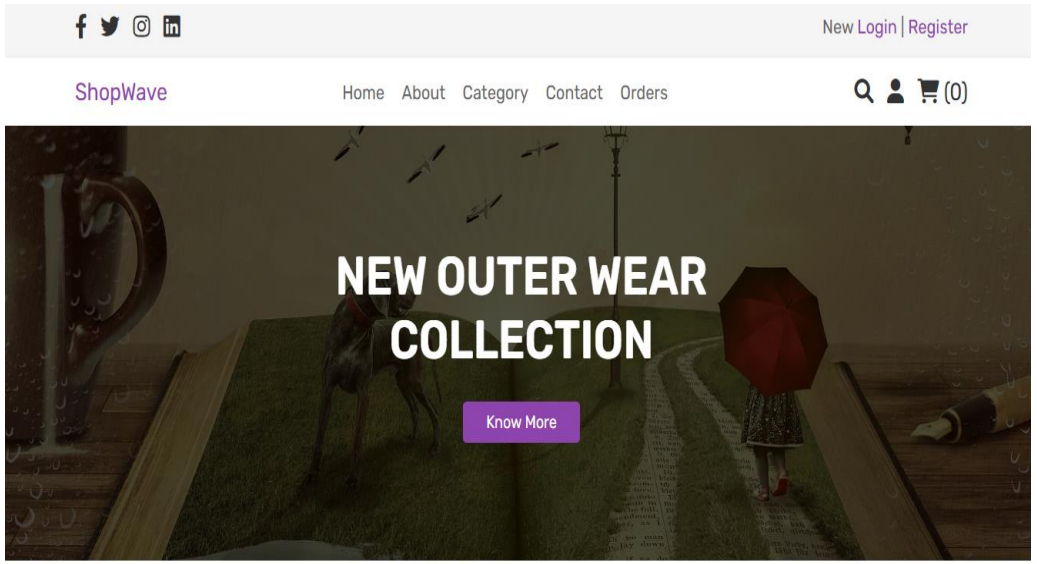
Shirt
$500/-
Update Delete

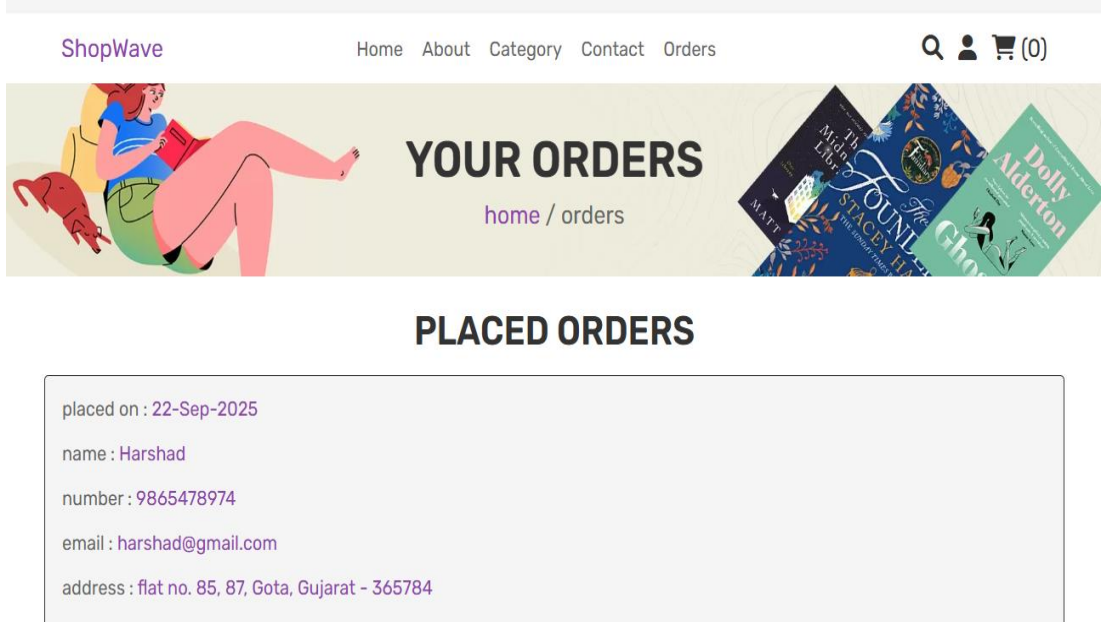


T-shirt
$450/-
Update Delete



Crop
$550/-
Update Delete


New Login | Register
ShopWave
Home About Category Contact Orders
(0)
NEW OUTER WEAR
COLLECTION
Know More


ShopWave
Home About Category Contact Orders
(0)
YOUR ORDERS
home / orders
PLACED ORDERS
placed on : 22-Sep-2025
name : Harshad
number : 9865478974
email : harshad@gmail.com
address : flat no. 85, 87, Gota, Gujarat - 365784

Thank You